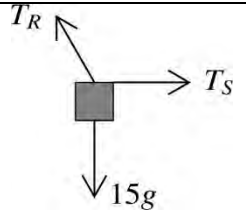
13	**(a)**			**B1** **[1]**	**1.1a**	Three forces in approximately correct directions, with arrows and labels; tensions must be distinct Accept W or mg for the weight; condone missing g for this mark	
13	**(b)**		Resolve vertically: $T_R \cos 30° = 15g$ $T_R = \dfrac{15 \times 9.8}{\cos 30°} = 98\sqrt{3} = 170$ N (3sf)	**M1** **A1** **[2]**	**3.3** **1.1b**	Forming equilibrium equation (allow sin/cos interchange) Oe	
13	**(c)**		Resolve horizontally: $T_S = T_R \sin 30°$ $T_S = 84.9$ N (3sf)	**M1** **A1** **[2]**	**1.1a** **1.1b**	Allow sin/cos interchange if consistent with (b) Oe	Accept $T_S = 15g \tan 30°$